Siqi Yang

Ph.D. Candidate, Institute for Artificial Intelligence, Peking University

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Research Interests

Neuromorphic Imaging (Spike Camera), Inverse Rendering & Relighting, Video Generation & World Models

Education

09/2022 - Present | **Peking University**, Beijing, China
Ph.D. candidate in Computer Science, Camera Intelligence Lab
Advised by Prof. Boxin Shi
Expected graduation: July 2027

09/2018 - 07/2022 | **Peking University**, Beijing, China
B.S. in Computer Science, Turing Honor Class

Experience

12/2025 - Present | **Shanda AI Research Tokyo**, Tokyo, Japan
Research Intern
Multi-player world models, panorama video generation, generative game platform

05/2019 - 08/2019 | **Hyperplane Lab, Peking University**, Beijing, China
Research Intern
Convolutional neural network quantization and pruning

07/2018 - 08/2018 | **Youtu X-Lab, Tencent**, Shenzhen, China
Research Intern
Object detection and classification on the Tencent Weishi dataset

Awards and Honors

2025 | **PKU-Huawei Turing Graduate Program Scholarship**

2025 | **Huawei Hackathon, Software Challenge**, Gold Medal

2019 | **Peking University Academic Excellence Award**

2018 | **China Collegiate Programming Contest (CCPC), Guilin**, Gold Medal

2017 | **National Olympiad in Informatics (NOI)**, Silver Medal

2016 – 2017 | **Other Informatics Olympiads**, NOIP First Prize (×2); APIO, CTSC, WC Bronze Medals

Academic Service

Conference Reviewer: CVPR · ICCV · ECCV · NeurIPS · SIGGRAPH · ACCV · PRCV · ICIG

Journal Reviewer: TPAMI

Teaching

2021 & 2022 | **Algorithm Design and Analysis**
Teaching Assistant, Peking University

2020 | **Introduction to Computing**
Teaching Assistant, Peking University

2017 - 2020 | **Informatics Olympiad (NOI/NOIP) Training**
Guest Lecturer, training camps in Beijing, Jinan, Qingdao, and Shijiazhuang

Publications

** denotes equal contribution.*

- **Siqi Yang**, Yimu Wang, Ziqi Cai, Chengxuan Zhu, Shuchen Weng, Erwin Wu, Zhaofei Yu, Kaipeng Zhang, Boxin Shi. PAWA: Panoramic Action-conditioned World Modeling for Multiple Agents. *arXiv preprint*, 2026.
- Yunheng Liu, Ziqi Cai, **Siqi Yang**, Yimu Wang, Minggui Teng, Jiaming Tan, Shuchen Weng, Erwin Wu, Kaipeng Zhang, Boxin Shi. SPW-Nav: A Streaming Panoramic World Model for Language-Guided Navigation. *arXiv preprint*, 2026.
- Zheng-Hui Huang, Guixu Lin, Jiacheng Lin, Yi-Chuan Huang, Ruihan Yu, Muyao Niu, **Siqi Yang**, Yu-Lun Liu, Yung-Yu Chuang, Kaipeng Zhang, Zhixiang Wang. Programmable World Model. *arXiv preprint*, 2026.
- Ziqi Cai, **Siqi Yang**, Yimu Wang, Zixian Gao, Yunheng Liu, Shuchen Weng, Erwin Wu, Kaipeng Zhang, Boxin Shi. MASS: Multiplayer World Models with Authoritative Shared State. *arXiv preprint*, 2026.
- Guixu Lin, Zheng-Hui Huang, **Siqi Yang**, Ming-Hsuan Yang, Kaipeng Zhang, Zhixiang Wang. Generative World Renderer at the Speed of Play. *arXiv preprint*, 2026.
- Chu Zhou*, **Siqi Yang***, Kailong Zhang, Heng Guo, Zhaofei Yu, Boxin Shi, Imari Sato. High-Speed Full-Color HDR Imaging via Unwrapping Modulo-Encoded Spike Streams. *arXiv preprint*, 2026.
- Haojie Zheng, Yixin Yang, **Siqi Yang**, Shuchen Weng, Boxin Shi. InstructAV2AV: Instruction-Guided Audio-Video Joint Editing. *SIGGRAPH Asia*, 2026 (to appear).
- Jiaming Tan, Zhen Li, Shuwei Shi, Minggui Teng, **Siqi Yang**, Yuwei Wu, Bo Zheng, Chuanhao Li, Kaipeng Zhang. MUGEN: Interactive Panoramic World Exploration via Camera Control. *SIGGRAPH Asia*, 2026 (to appear).
- Yakun Chang, Zhaojun Huang, **Siqi Yang**, Yeliduosi Xiaokaiti, Shikui Wei, Yao Zhao, Tiejun Huang, Boxin Shi. HFR and HDR Video from Multi-Attenuated Spikes Using a Rapidly Rotating SpokeND Filter. *CVPR*, 2026.
- Haojie Zheng, Shuchen Weng, Jingqi Liu, **Siqi Yang**, Boxin Shi, Xinlong Wang. Audio-sync Video Instance Editing with Granularity-Aware Mask Refiner. *CVPR*, 2026.
- Yifei Xia, Shuchen Weng, **Siqi Yang**, Jingqi Liu, Chengxuan Zhu, Minggui Teng, Zijian Jia, Han Jiang, Boxin Shi. PanoWan: Lifting Diffusion Video Generation Models to 360° with Latitude/Longitude-aware Mechanisms. *NeurIPS*, 2025.
- **Siqi Yang**, Jinxiu Liang, Zhaojun Huang, Yeliduosi Xiaokaiti, Yakun Chang, Zhaofei Yu, Boxin Shi. SpikeDiff: Zero-shot High-Quality Video Reconstruction from Chromatic Spike Camera and Sub-Millisecond Spike Streams. *ICCV*, 2025.
- Jinxiu Liang, Bohan Yu, **Siqi Yang**, Haotian Zhuang, Jieji Ren, Peiqi Duan, Boxin Shi. EventUPS: Uncalibrated Photometric Stereo Using an Event Camera. *ICCV*, 2025.
- **Siqi Yang***, Zhaojun Huang*, Yakun Chang, Bin Fan, Zhaofei Yu, Boxin Shi. Real-Data-Driven 2000 FPS Color Video from Mosaicked Chromatic Spikes. *ECCV*, 2024.
- **Siqi Yang**. Focal Stack and Light Field Imaging (Chapter 5). In Computational Photography: Imaging Model Theory and Deep Learning Practice, Boxin Shi (Ed.), *China Machine Press*, 2024.
- **Siqi Yang***, Xuanning Cui*, Yongjie Zhu, Jiajun Tang, Si Li, Zhaofei Yu, Boxin Shi. Complementary Intrinsics From Neural Radiance Fields and CNNs for Outdoor Scene Relighting. *CVPR*, 2023.
- Bohan Yu, **Siqi Yang**, Xuanning Cui, Siyan Dong, Baoquan Chen, Boxin Shi. MILO: Multi-bounce Inverse Rendering for Indoor Scene with Light-emitting Objects. *TPAMI*, 2023.

Patents

- Method and Apparatus for Self-Calibrating Surface Normal Field Reconstruction Based on Asynchronous Event Streams. China patent application, pending.

Skills

Programming: Python · C++ · CUDA · Rust · Lisp · Nix

Deep Learning: PyTorch · JAX · TensorFlow (v1 & v2) · TensorLayer · Keras

Graphics & Imaging: NeRF · 3DGS · Spike sensor pipeline · Differentiable rendering

Infrastructure: Docker · Ray · Cloudflare Workers · GCP · Git

Languages: Chinese · English

杨思祺

北京大学 人工智能研究院 博士研究生

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研究方向

神经形态成像（脉冲相机）、逆向渲染与重光照、视频生成与世界模型

教育经历

2022.09 - 至今 | **北京大学**，北京
计算机科学 博士研究生，相机智能实验室
导师：施柏鑫 教授
预计 *2027* 年 *7* 月毕业

2018.09 - 2022.07 | **北京大学**，北京
计算机科学 学士，图灵班

实习经历

2025.12 - 至今 | **Shanda AI Research Tokyo**，东京
研究实习生
多人世界模型、全景视频生成、生成式游戏平台

2019.05 - 2019.08 | **Hyperplane Lab**，北京大学，北京
研究实习生
卷积神经网络量化与剪枝

2018.07 - 2018.08 | **腾讯优图 X-Lab**，深圳
研究实习生
微视数据集上的目标检测与分类

荣誉奖项

2025 | **北大-华为图灵计划研究生奖学金**

2025 | **华为黑客松 软件挑战赛，金奖**

2019 | **北京大学学习优秀奖**

2018 | **中国大学生程序设计竞赛（CCPC）桂林站，金牌**

2017 | **全国青少年信息学奥林匹克竞赛（NOI），银牌**

2016 – 2017 | **其他信息学竞赛**，NOIP 一等奖（×2）；APIO、CTSC、WC 铜牌

学术服务

会议审稿人： CVPR · ICCV · ECCV · NeurIPS · SIGGRAPH · ACCV · PRCV · ICIG

期刊审稿人： TPAMI

教学经历

2021 & 2022 | **算法设计与分析**
北京大学 助教

2020 | **计算概论**
北京大学 助教

2017 - 2020 | **信息学竞赛培训**
客座讲师，北京、济南、青岛、石家庄

学术论文

* 表示共同一作

- **Siqi Yang**, Yimu Wang, Ziqi Cai, Chengxuan Zhu, Shuchen Weng, Erwin Wu, Zhaofei Yu, Kaipeng Zhang, Boxin Shi. PAWA: Panoramic Action-conditioned World Modeling for Multiple Agents. *arXiv preprint*, 2026.
- Yunheng Liu, Ziqi Cai, **Siqi Yang**, Yimu Wang, Minggui Teng, Jiaming Tan, Shuchen Weng, Erwin Wu, Kaipeng Zhang, Boxin Shi. SPW-Nav: A Streaming Panoramic World Model for Language-Guided Navigation. *arXiv preprint*, 2026.
- Zheng-Hui Huang, Guixu Lin, Jiacheng Lin, Yi-Chuan Huang, Ruihan Yu, Muyao Niu, **Siqi Yang**, Yu-Lun Liu, Yung-Yu Chuang, Kaipeng Zhang, Zhixiang Wang. Programmable World Model. *arXiv preprint*, 2026.
- Ziqi Cai, **Siqi Yang**, Yimu Wang, Zixian Gao, Yunheng Liu, Shuchen Weng, Erwin Wu, Kaipeng Zhang, Boxin Shi. MASS: Multiplayer World Models with Authoritative Shared State. *arXiv preprint*, 2026.
- Guixu Lin, Zheng-Hui Huang, **Siqi Yang**, Ming-Hsuan Yang, Kaipeng Zhang, Zhixiang Wang. Generative World Renderer at the Speed of Play. *arXiv preprint*, 2026.
- Chu Zhou*, **Siqi Yang***, Kailong Zhang, Heng Guo, Zhaofei Yu, Boxin Shi, Imari Sato. High-Speed Full-Color HDR Imaging via Unwrapping Modulo-Encoded Spike Streams. *arXiv preprint*, 2026.
- Haojie Zheng, Yixin Yang, **Siqi Yang**, Shuchen Weng, Boxin Shi. InstructAV2AV: Instruction-Guided Audio-Video Joint Editing. *SIGGRAPH Asia*, 2026 (to appear).
- Jiaming Tan, Zhen Li, Shuwei Shi, Minggui Teng, **Siqi Yang**, Yuwei Wu, Bo Zheng, Chuanhao Li, Kaipeng Zhang. MUGEN: Interactive Panoramic World Exploration via Camera Control. *SIGGRAPH Asia*, 2026 (to appear).
- Yakun Chang, Zhaojun Huang, **Siqi Yang**, Yeliduosi Xiaokaiti, Shikui Wei, Yao Zhao, Tiejun Huang, Boxin Shi. HFR and HDR Video from Multi-Attenuated Spikes Using a Rapidly Rotating SpokeND Filter. *CVPR*, 2026.
- Haojie Zheng, Shuchen Weng, Jingqi Liu, **Siqi Yang**, Boxin Shi, Xinlong Wang. Audio-sync Video Instance Editing with Granularity-Aware Mask Refiner. *CVPR*, 2026.
- Yifei Xia, Shuchen Weng, **Siqi Yang**, Jingqi Liu, Chengxuan Zhu, Minggui Teng, Zijian Jia, Han Jiang, Boxin Shi. PanoWan: Lifting Diffusion Video Generation Models to 360° with Latitude/Longitude-aware Mechanisms. *NeurIPS*, 2025.
- **Siqi Yang**, Jinxiu Liang, Zhaojun Huang, Yeliduosi Xiaokaiti, Yakun Chang, Zhaofei Yu, Boxin Shi. SpikeDiff: Zero-shot High-Quality Video Reconstruction from Chromatic Spike Camera and Sub-Millisecond Spike Streams. *ICCV*, 2025.
- Jinxiu Liang, Bohan Yu, **Siqi Yang**, Haotian Zhuang, Jieji Ren, Peiqi Duan, Boxin Shi. EventUPS: Uncalibrated Photometric Stereo Using an Event Camera. *ICCV*, 2025.
- **Siqi Yang***, Zhaojun Huang*, Yakun Chang, Bin Fan, Zhaofei Yu, Boxin Shi. Real-Data-Driven 2000 FPS Color Video from Mosaicked Chromatic Spikes. *ECCV*, 2024.
- 杨思祺。《计算摄像学:成像模型理论与深度学习实践》第 5 章「焦点堆栈与光场摄像」。施柏鑫 主编，机械工业出版社，2024。
- **Siqi Yang***, Xuanning Cui*, Yongjie Zhu, Jiajun Tang, Si Li, Zhaofei Yu, Boxin Shi. Complementary Intrinsics From Neural Radiance Fields and CNNs for Outdoor Scene Relighting. *CVPR*, 2023.
- Bohan Yu, **Siqi Yang**, Xuanning Cui, Siyan Dong, Baoquan Chen, Boxin Shi. MILO: Multi-bounce Inverse Rendering for Indoor Scene with Light-emitting Objects. *TPAMI*, 2023.

专利

- 一种基于异步事件流的自校准表面法向场重建方法及装置。中国发明专利，已提交申请。

技能

编程语言： Python · C++ · CUDA · Rust · Lisp · Nix

深度学习： PyTorch · JAX · TensorFlow (v1 & v2) · TensorLayer · Keras

图形与成像： NeRF · 3DGS · Spike sensor pipeline · Differentiable rendering

基础设施： Docker · Ray · Cloudflare Workers · GCP · Git

语言： 中文 · 英文